

Maximum Likelihood Estimation

State-space models, filtering and DSGE likelihoods

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SESSION OBJECTIVE

- ▶ Express a solved DSGE model as a state-space system.
- ▶ Evaluate the likelihood with the Kalman filter.
- ▶ Interpret estimates alongside identification and diagnostics.

Handout: Session 3 **Code:** `code/session3/`

PLAN

① State-space representation

② Likelihood

③ Estimation

④ Takeaways

FROM MODEL SOLUTION TO OBSERVATIONS

$$\underbrace{s_t}_{\text{state}} = Ts_{t-1} + R\varepsilon_t, \quad \underbrace{y_t}_{\text{data}} = Zs_t + d + \eta_t.$$

- ▶ The transition equation propagates latent states.
- ▶ The measurement equation links the model to transformed data.
- ▶ Measurement error η_t is an economic and statistical assumption.

DATA TRANSFORMATIONS ARE PART OF THE MEASUREMENT EQUATION



Always record: provider, dataset, series, frequency, sample, missing values and transformation.

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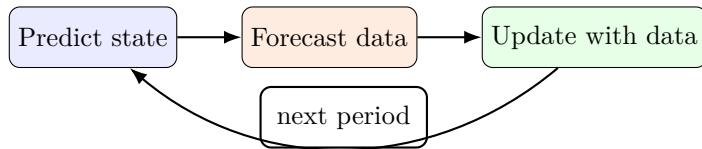
THE LIKELIHOOD DECOMPOSES PREDICTION ERRORS

At each date, the Kalman filter delivers a forecast error v_t and covariance F_t :

$$\log L(\theta) = -\frac{1}{2} \sum_t [\log |F_t| + v_t' F_t^{-1} v_t + n_y \log(2\pi)] .$$

- ▶ Parameters affect both the forecast and its uncertainty.
- ▶ Missing observations can be handled systematically.
- ▶ The filter turns a dynamic model into an evaluable criterion.

KALMAN FILTER: INTUITION



The gain balances model-implied uncertainty against measurement uncertainty.

PLAN

- 1 State-space representation
- 2 Likelihood
- 3 Estimation**
- 4 Takeaways

MAXIMUM LIKELIHOOD WORKFLOW

- 1 Specify model, observables and parameter domain.
- 2 Build the state-space solution and evaluate $\log L(\theta)$.
- 3 Optimize from multiple informed starting values.
- 4 Inspect convergence, curvature, residuals and economic plausibility.

A NUMERICAL OPTIMUM IS NOT THE END

Check

- ▶ optimization status
- ▶ parameter bounds
- ▶ Hessian / standard errors
- ▶ forecast errors

Question

Is the likelihood informative about this parameter, or is the result driven by a restriction?

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TAKEAWAYS AND PREPARATION

- ▶ Estimation begins with the measurement equation, not the optimizer.
- ▶ The Kalman filter evaluates likelihoods recursively.
- ▶ Diagnostics distinguish a computational answer from credible inference.

Before Session 4: inspect the observables and prior choices in `RBC_estim.mod`.